

Completing the square and the discriminant

Cambridge insert: the form that hands you the vertex, and the number that counts the roots before you find them.

LESSON 4–5

2 × 40 MIN

ALGEBRA 10, REVISION

P1 · 1.3, 1.6

LESSON CLOCK

5'

14'

12'

5'

4'

Expand three squares to see the pattern backwards 5 min

Completing the square 14 min

What the form tells you 12 min

The discriminant 5 min

Homework 4 min

LEARNING OBJECTIVES

- Write $x^2 + bx + c$ in the form $(x + p)^2 + q$.
- Read the vertex and the least value off that form.
- Use D to say how many real roots there are.
- Find the values of a parameter that make a quadratic touch the axis.

TEXTBOOK

National Повторение, pp. 6–13

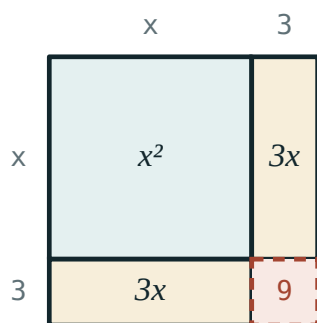
Cambridge Pure Mathematics 1, pp. 7–16

Workbook P1 Exercise 1B, 1F

01 Explanation

Completing the square

Expanding $(x + 3)^2 = x^2 + 6x + 9$ shows the pattern: the number inside is **half** the coefficient of x , and the constant is its square. Run it backwards and you can turn any $x^2 + bx$ into a square, at the cost of one correction:



$$x^2 + 6x + 9 = (x + 3)^2$$

$$\begin{aligned} \text{so } x^2 + 6x \\ = (x + 3)^2 - 9 \end{aligned}$$

$x^2 + 6x$ is one small square short of $(x + 3)^2$. Add the missing 9 — and take it away again.

$$x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$$

So $x^2 + 6x + 1 = (x + 3)^2 - 9 + 1 = (x + 3)^2 - 8$.

WHEN $A \neq 1$

Take a out of the first two terms first: $2x^2 + 8x + 3 = 2(x^2 + 4x) + 3 = 2[(x + 2)^2 - 4] + 3 = 2(x + 2)^2 - 5$. The bracket must be multiplied back out.

What the completed form gives you free

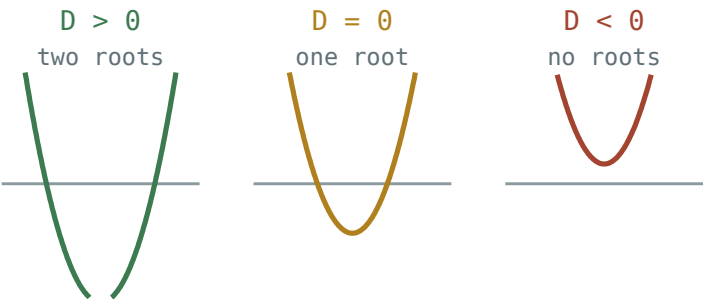
From $y = (x + p)^2 + q$ you can read, without any further work:

, at $x = -p$ — real only if $q \leq 0$	
QUESTION	ANSWER
the vertex	$(-p, q)$
the axis of symmetry	$x = -p$
the least value of y	q
the range	$y \geq q$
the roots	$(x + p)^2 = -q$

A square is never negative, so $(x + p)^2 \geq 0$ and therefore $y \geq q$ always. That one line answers every "find the minimum value" question in the quarter.

The discriminant

$$D = b^2 - 4ac$$



$D > 0$ cuts the axis twice, $D = 0$ touches it, $D < 0$ misses it entirely.

D	ROOTS	THE PARABOLA
$D > 0$	two different real roots	cuts the x -axis twice
$D = 0$	one repeated root	touches the axis — it is a tangent
$D < 0$	no real roots	misses the axis completely

“Touches the axis”, “has a repeated root”, “is a tangent to the x -axis” and “has exactly one solution” are four ways of writing $D = 0$. Recognising them is most of the marks in this topic.

02 Worked examples

EXAMPLE 1 Write $x^2 - 8x + 21$ in completed-square form and state its least value.

- 1 half of -8 is -4
- 2 $x^2 - 8x = (x - 4)^2 - 16$
- 3 $(x - 4)^2 - 16 + 21 = (x - 4)^2 + 5$
- 4 Least value 5, at $x = 4$.
A square is at least 0.

ANSWER

$(x - 4)^2 + 5$; least value 5

EXAMPLE 2 Write $3x^2 + 12x + 7$ in the form $a(x + p)^2 + q$.

① $3(x^2 + 4x) + 7$

Take the 3 out of the first two terms only.

② $x^2 + 4x = (x + 2)^2 - 4$

③ $3[(x + 2)^2 - 4] + 7$

④ $3(x + 2)^2 - 12 + 7 = 3(x + 2)^2 - 5$

Multiply the 3 through.

ANSWER

$$3(x + 2)^2 - 5$$

EXAMPLE 3 Find the values of k for which $x^2 + kx + 9 = 0$ has a repeated root.

① A repeated root means $D = 0$.

② $D = k^2 - 4 \cdot 1 \cdot 9 = k^2 - 36$

③ $k^2 - 36 = 0$

④ $k = \pm 6$

Both work: $(x \pm 3)^2$.

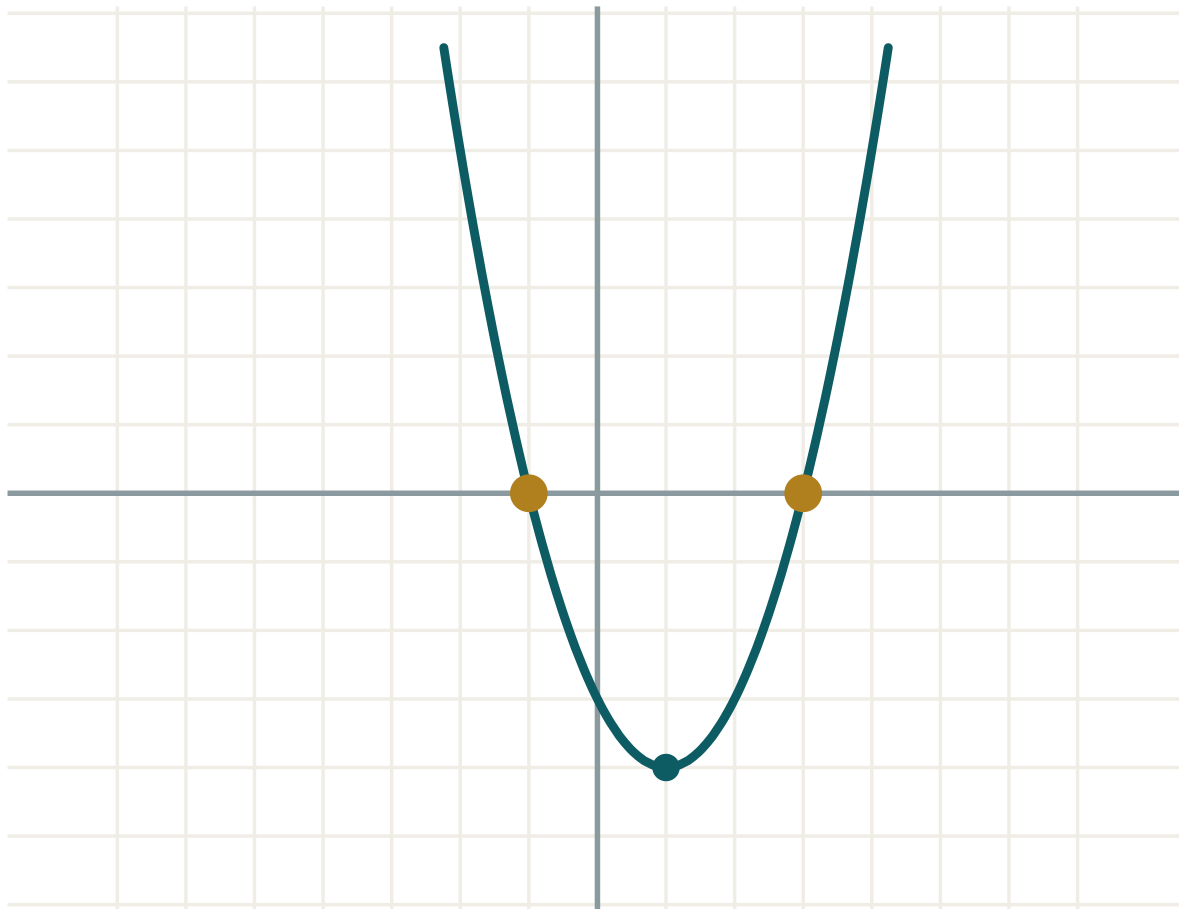
ANSWER

$$k = 6 \text{ or } k = -6$$

03 Interactive model

Move the coefficients and watch D change sign as the parabola lifts off the axis.

The discriminant in action



a 1

b -2

c -3

 $D = b^2 - 4ac$ 16

roots two

 x_1 -1 x_2 3 $x_1 + x_2$ 2 $x_1 \cdot x_2$ -3

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Completing the square	To'la kvadratga keltirish	Выделение полного квадрата
Perfect square	To'la kvadrat	Полный квадрат
Vertex	Uchi	Вершина
Minimum value	Eng kichik qiymat	Наименьшее значение
Maximum value	Eng katta qiymat	Наибольшее значение
Discriminant	Diskriminant	Дискриминант
Repeated root	Karrali ildiz	Кратный корень
Tangent to the axis	O'qqa urinma	Касается оси
Axis of symmetry	Simmetriya o'qi	Ось симметрии

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$x^2 + 10x$ completed is:

The least value of $(x - 2)^2 + 7$ is:

If $D = 0$ the parabola:

The vertex of $y = (x + 4)^2 - 3$ is:

$(4, -3)$

$(-4, -3)$

$(-4, 3)$

$(4, 3)$

$x^2 + 2x + 5 = 0$ has:

two roots

one root

no real roots

infinitely many

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Complete the square: $x^2 + 4x$

ANSWER $(x + 2)^2 - 4$

2. Complete the square: $x^2 - 6x$

ANSWER $(x - 3)^2 - 9$

3. Complete the square: $x^2 + 2x + 5$

ANSWER $(x + 1)^2 + 4$

4. State the vertex of $y = (x - 1)^2 + 2$

ANSWER $(1, 2)$

5. Find D for $x^2 + 6x + 9 = 0$

ANSWER 0

6. Find D for $x^2 + x + 1 = 0$

ANSWER -3

7. State the least value of $(x + 3)^2 - 1$

ANSWER -1

Medium · the standard question

7 PROBLEMS

1. Complete the square: $x^2 - 10x + 30$

ANSWER $(x - 5)^2 + 5$

2. Complete the square: $2x^2 + 8x + 1$

ANSWER $2(x + 2)^2 - 7$

3. Find the vertex of $y = x^2 - 4x + 7$

ANSWER $(2, 3)$

4. Find k so that $x^2 + kx + 25 = 0$ has one root.

ANSWER $k = \pm 10$

5. For what k does $x^2 + 4x + k = 0$ have two roots?

ANSWER $k < 4$

6. Find the least value of $3x^2 - 12x + 5$

ANSWER $3(x - 2)^2 - 7$, least -7

7. Show $x^2 - 2x + 6$ is always positive.

ANSWER $(x - 1)^2 + 5 \geq 5 > 0$

Hard · stretch and reason

7 PROBLEMS

1. Complete the square: $5 - 4x - x^2$

ANSWER $9 - (x + 2)^2$, greatest value 9

2. For what k is $y = x^2 + kx + k + 3$ a tangent to the x -axis?

ANSWER $k^2 - 4k - 12 = 0$, so $k = 6$ or $k = -2$

3. Find the range of $y = 2x^2 - 8x + 11$

ANSWER $2(x - 2)^2 + 3$, so $y \geq 3$

4. The line $y = mx + 1$ is a tangent to $y = x^2 + 3$. Find m .

ANSWER $x^2 - mx + 2 = 0$, $D = m^2 - 8 = 0$, $m = \pm 2\sqrt{2}$

5. For what values of k does $kx^2 + 4x + 1 = 0$ have no real root?

ANSWER $16 - 4k < 0$, so $k > 4$

6. Prove $x^2 + bx + c$ has a least value of $c - b^2/4$.

ANSWER $(x + \frac{b}{2})^2 + c - \frac{b^2}{4}$; the square is at least 0.

7. Find the greatest value of $\frac{1}{x^2 - 4x + 8}$

ANSWER the denominator is least at 4, so the greatest value is $\frac{1}{4}$

07 Homework

Homework — 6 problems

P1 Exercise 1B and 1F. Always state the vertex once you have the completed form.

- Complete the square: $x^2 + 12x + 40$
- Complete the square: $2x^2 - 12x + 5$
- Find the vertex and least value of $y = x^2 - 6x + 11$
- Find k so that $x^2 + kx + 16 = 0$ has a repeated root.

5. *For what k does $x^2 + 6x + k = 0$ have no real root?*

6. *Show that $x^2 + 4x + 9$ is positive for every x .*